PROJECT DOCUMENTATION

Team ID:	SWTID-2026-7788
Project Title:	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Submission Date:	29 September 2026

1. Introduction

- **Project Title:** Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
- **Team Members:** SWTID-2026-7788 Team Members (Workflow Automation & Platform Specialists)

2. Project Overview

- **Purpose:** To automate and streamline standard laptop procurement and fulfillment processes using ServiceNow Flow Designer, eliminating manual approval bottlenecks and ensuring transparent tracking.
- **Features:** Automated catalog item requests, dynamic approval routing based on department hierarchy, real-time status tracking, and automated task assignment.

3. Architecture

- **Frontend:** ServiceNow Service Portal providing a seamless user interface for requesting standard hardware.
- **Backend:** ServiceNow Flow Designer engine, Script Includes, and Business Rules executing server-side automation logic.
- **Database:** ServiceNow CMDB and custom Procurement tables storing request states, audit logs, and asset mappings.

4. Setup Instructions

- **Prerequisites:** ServiceNow Developer Instance, Flow Designer plugin enabled, Service Catalog activated.
- **Installation:** Step-by-step guide to import update sets containing catalog items, workflow actions, and custom form configurations into the target instance.

5. Folder Structure

- **Client:** Service Catalog widgets, client-side validation scripts, and portal branding configurations.
- **Server:** Flow Designer flows, subflows, custom actions, Script Includes, and scheduled job scripts.

6. Running the Application

- Provide commands or execution steps to trigger and test the automated workflows locally.
- **Initiation:** Navigate to Service Portal -> Order Standard Laptop.
- **Execution:** Submit the request to automatically trigger the Flow Designer automated workflow pipeline.

7. API Documentation

- Document all endpoints exposed by the backend for external procurement and vendor integration.
- Include request methods, parameters, and example responses for REST endpoints like `/api/now/table/sc_req_item`.

8. Authentication

- Explain how authentication and authorization are handled in the project using Role-Based Access Control (RBAC) and OAuth 2.0 integration for external API requests.

9. User Interface

- Provide screenshots or GIFs showcasing Service Catalog request forms, Flow Designer execution trace dashboards, and manager approval portal views.

10. Testing

- Describe the testing strategy and tools used, including unit testing of custom scripts, integration testing of catalog triggers, and UAT execution.

11. Screenshots or Demo

- Provide screenshots or a link to a demo showcasing workflow execution logs, catalog item UI layout, and approval notification email templates.

12. Known Issues

- Document any known bugs or issues, such as strict dependency on predefined hardware catalog options and department approval matrix data.

13. Future Enhancements

- Outline potential future features, such as AI-powered predictive inventory forecasting and direct API integration with third-party laptop vendor drop-shipping systems.